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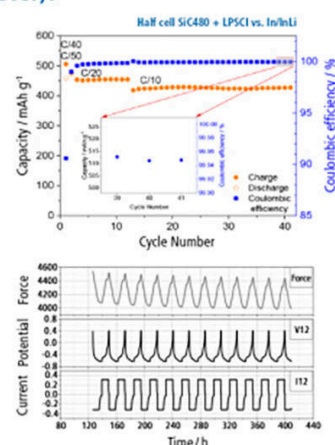
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Investigation of the ripeness of oil palm fresh fruit bunches using bio-speckle imaging

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Abstract. The ripeness of the oil palm Fresh Fruit Bunches (FFB) determines the yield of the oil produced. Traditionally there are two ways to determine FFB ripeness which are the number of loose fruits and the color changes. Nevertheless, one drawback of visual determination is subjective and qualitative judgment. In this study, the FFB ripeness was investigated using laser based image processing technique. The advantages of using this technique are non-destructive, simple and quantitative. The working principle of the investigation is that a FFB is inserted into a light tight box which contains a laser diode and a CMOS camera, the FFB is illuminated, and then an image is recorded. The FFB image recorder was performed on four FFB fractions i.e. F0, F3, F4 and F5 on the front and rear surfaces at three sections. The recorded images are speckled granules that have light intensity variation (bio-speckle imaging). The feature extracted from the specked image is the contrast value obtained from the average gray value intensity and the standard deviation. Based on the contrast values, the four fractions of FFB can be grouped into three levels of ripeness of unripe (F0), ripe (F3) and overripe (F4 and F5) on the front surface of base section of FFB by 75%.

1. Introduction

In the last few decades the problem of determining the maturity and ripeness level of oil palm fresh fruit bunches (FFB) has become a popular research topic in the palm oil industry. Indonesia, which is one of the largest Crude Palm Oil producing and exporting countries in the world, makes palm oil the main source of foreign exchange in Indonesia. Fruit maturity and ripeness determine the yield of oil produced [1], therefore the determination of FFB maturity and ripeness is one of the problems for farmers and oil palm companies in Indonesia. The determination of FFB ripeness before harvested or in sorting and grading process is still done manually or traditionally using the vision (human vision) by trained and experienced workers. Several studies on FFB ripeness have been conducted in the last decade, one of which is a review of several techniques for predicting maturity of oil palm fruit based on visual observation, oil content and image processing [2].

FFB ripeness is classified by several standards. In general, there are three categories of FFB ripeness those are unripe, ripe, and overripe. Traditionally there are two ways to determine the ripeness stages of FFB that are base on the number of fruits that detached and fell from a FFB and based on the color of FFB. However, one drawback of visual determination is subjective and qualitative judgment.

This research investigates FFB ripeness categories using a laser based image processing technique. In order to get the image, a FFB is inserted into a light tight box containing a 650 nm laser diodes and a CMOS camera. The diode laser illuminates the FFB and the FFB speckle image is recorded using the



camera. The resulting image has a light and dark speckle pattern. In the field of biological science, the activity of biological matter such as a fruit which is irradiated by a laser beam is called Bio-speckle. Bio-speckle imaging and image processing techniques are potential combination to analyze and classified ripeness of oil palm FFBs.

2. Literature Review

The use of laser light in agriculture become intensive in last decade because it is non-destructive, simple, and fast. The optical method is potential to substitute or accompany laboratory test methods that are expensive, invasive, and time consuming. Laser Speckle Imaging (LSI) is an optical method that can be used to detect the level of ripeness and abnormalities in the fruit. This technique relies on the interaction of light and the fruit material and monochromatic light. When laser light illuminates the fruit surface, light can be absorbed, reflected, transmitted, and scattered by the surface of the material. Fruit has several physical properties which are represented by physical parameters such as density, porosity, surface roughness, diameter, and color. Light that concerns uneven surfaces will experience two types of reflections that are specular reflections and diffuse reflections. The reflected light beams will have different phases that will cause constructive and destructive interference resulting in some light and dark speckle granules. The method of speckling pattern caused by biological material such as fruit is called Bio-speckle. The bio-speckle phenomenon occurs when biological matter is irradiated by coherent light (laser light). The diffused light from surface roughness is optically recorded and produces the image with light and dark speckles of light [3].

There are two type of speckle pattern: static and dynamic. The static pattern is derived from the static part of the fruit network, while the dynamic pattern comes from particles or molecules that move in the tissues i.e in the process of maturity or ripeness. The dynamics of this speckle pattern can be statistically analyzed. This method is based on variations of laser light intensity or contrast dissipated by biological samples. Fruit is a complex element consisting of molecules of fruit tissue built of various sizes depending on the level of maturity and ripeness, These molecular sizes will affect the optical properties of the fruit associated with the skin's ability and the fruit tissue to absorb and scatter light so that it can be used to monitor fruit maturity levels [4].

The efforts of finding nondestructive methods for the detection of fruits for both pre- and post-harvest conditions have been widely practiced. The nondestructive method for estimating mango maturity uses color as one of the characteristics of maturity [5]. In addition, the use of image processing algorithm has also been developed for sorting and grading in fruits and vegetables [6]. A study of the determination of FFB maturity and ripeness has also been done and many methods have been developed. Identification of nondestructive maturity generally uses computer vision and image processing techniques. The characteristic of the extraction is the color of the image. The CCD camera recording is then processed using software to classify FFB maturity [7]. This technique provides classification accuracy between 83.5% - 94%. In addition, a grading system using a computer-based photogrammetric method, has been conducted in which the FFB is illuminated by a homogeneous light bulb in a box [8]. Furthermore, the ripeness category is determined from the Color Digital Number (DN). Automatic FFB grading systems have also been developed in estimating the ripeness fraction, oil content and free fatty acids of FFB using two statistical methods of forward stepwise multiple linear regression and multilayer-perceptron artificial neural network analysis. The estimation accuracy of both statistical methods is about 90%[9].

3. Material and Methodology

3.1 Tools and materials

The tools used in this research are a diode laser, a CMOS sensor camera, biconcave lenses, image processing software, and data processing as well as other optical components. The sample material studied was FFB of varieties of Tenera (Lonsum clone). FFBs are selected by four maturity fractions

(representing three levels of maturity) i.e. F0 (unripe), F3 (ripe), F4 and F5 (overripe). The tools and materials used are presented in the following table

Table 1. Tool and materials

Tool and materials	Information
Oil palm FFB	Tenera Variety
Optical bench	Thorlabs
Diode laser 650 nm	Roithner
Monochrome CMOS camera	Imaging Source
Camera lens Kamera 1.4/12mm	Navitar
Biconcave lens $f = -50$ mm	Thorlabs
Laptop	Sony
IC Capture 2.4	Imaging Source
Matlab	http://mathworks.com

3.2 Data

The FFB samples used represented three levels of ripeness and were selected from trees under the age of eight years. The ripeness stages of harvested FFBs were determined by an experience harvester. The FFBs were then brought to Lab and cleaned. The measurements were completed in 24 hours after harvest. The image recording is performed on two FFB surfaces, the rear and front surfaces. The rear surface is the surface facing the palm tree while the front surface is facing sunlight. The surface of FFB is irradiated in three parts, namely the base, the middle and the tip, so that each FFB gets six specimen images. A specimen image of each FFB was analyzed using MATLAB software to obtain average gray value intensities and standard deviation values on all parts of each surface. Both values are computed to obtain the value of the contrast that represents the ripeness level of each FFB.

3.3 Method

To observe the bio-speckle activity at three levels of FFB ripeness, we have calculated the speckle modulation pattern expressed as contrast in the FFB image. The contrast equation is as follows:

$$C = \frac{\sigma I}{\langle I \rangle} \quad (1)$$

$$I = \frac{\sum f(x,y)}{n} \quad (2)$$

$$\sigma I = \sqrt{\frac{\sum (f(x,y) - I)^2}{n-1}} \quad (3)$$

where

C	= contrast
σI	= standard deviation
I	= average of grey value intensity
$f(x, y)$	= grey value on spatial coordinates
n	= the number of spatial coordinates

The average gray value intensity shows the intensity distribution of the gray level, the standard deviation shows the intensity of gray and the contrast intensity levels represents variations in gray level intensity distribution. Equation (1) becomes the characteristic of extraction algorithm in this study.

4. Results and Discussion

Image of speckled FFB captured uses a monochrome CMOS camera with 650 nm diode laser wavelength. Supporting components are camera lens, biconcave lens -50 mm. The camera lens is used to adjust the focus and camera aperture and bi-concave lenses are used to spread the laser beam. Biconcave lens arrangement focusehe laser beam only on 2-5 pieces on FFB fruits because

illuminating the entire surfaces of FFB will produce dark dominant speckles. FFS objects are irradiated alternately in three parts of the base, middle and ends of the front and back surfaces of FFB. Each fraction is represented by three FFB, so that for each part on the surface 12 images are produced. CMOS cameras are installed as far as 44.3 cm and perpendicular to FFB. The laser distance to FFB is 58.6 cm, and the distance of the laser to the camera is 17.5 cm. The distances between these components is kept constant, so the width of the beam of laser beams on the surface of the FFB remains constant. The system is built in black box with size 60 cm x 60 cm x 90 cm. The black box function is to eliminate outside light interference during the recording process. Illustration of illumination and recording of FFB and FFB biospeckle results is presented in the following figure.

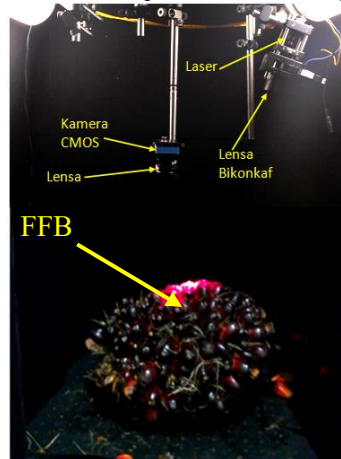


Figure 1. Radiation system and bio-speckle image recording

Image recording is done experimentally at different levels of FFB ripeness. The sample used was four fractions of FFB with three variations of ripeness level i.e. F0 (unripe), F3 (ripe), F4 and F5 (overripe). The speckled data retrieval is taken from the front and back surfaces of FFB. The fruits on the back surface of FFB have a variety of different shape and size. The backbone FFB base has many spikelet and many small fruits. Rear surface fruits have fewer gaps than the front surface. The front surface of FFB has uniform fruits on all parts. The front surface FFB tip has a smaller size than the rear end FFB tip.

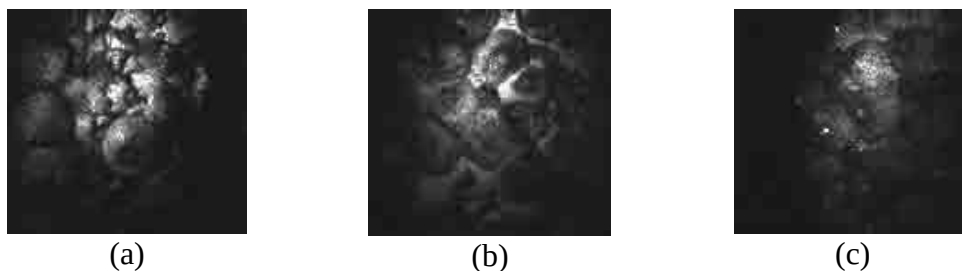


Figure 2. The pattern of the frontal specimen of FFB fraction F4 (a) base (b) middle (c) top

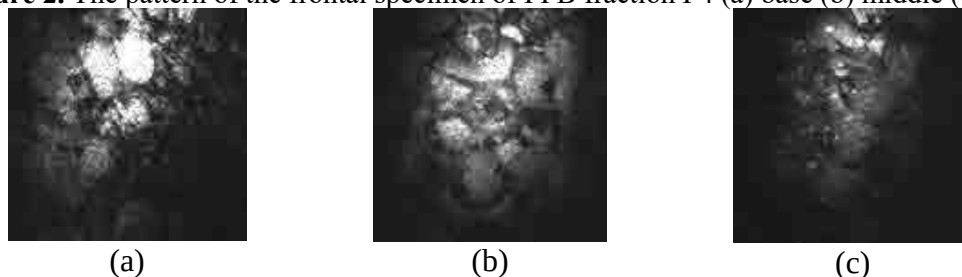


Figure 3. The pattern of the backside FFB specimen fraction F4 (a) base (b) middle (c) top

Speckle image data processing is done using MATLAB. Via this software, the average features of gray value and standard deviation intensity are extracted using equations (2) and (3). Then both values are entered into equation (1) to obtain a contrast value. Based on the observation of contrast value on 72 image data, it can be determined that the contrast value for $F0 \leq 1.4$, $F3 \leq 1.5$, $F4$ and $F5 > 1.5$. The results of the feature extraction, computation and category of FFB speckle patterns are presented in the following table.

Table 2. The results of feature extraction and category of front surface FFB

No	Label	(I)	(σI)	(C)	Category	Results	Remarks
1	F0S1 DP	9,2335	11,7414	1,2716	F0	True	DP:
2	F0S2 DP	9,1673	12,3662	1,3489	F0	True	Front
3	F0S3 DP	7,6634	10,5114	1,3716	F0	True	surface
4	F3S1 DP	11,2172	16,1706	1,4416	F3	True	of base
5	F3S2 DP	11,1350	16,0150	1,4383	F3	True	section
6	F3S3 DP	10,7062	15,9368	1,4886	F3	True	
7	F4S1 DP	16,0704	24,4794	1,5233	F4/F5	True	
8	F4S2 DP	12,0800	17,3159	1,4334	F3	False	
9	F4S3 DP	9,8844	13,1788	1,3333	F0	False	
10	F5S1 DP	12,0197	18,4556	1,5354	F4/F5	True	
11	F5S2 DP	12,0103	17,2111	1,4330	F3	False	
12	F5S3 DP	17,3642	27,2451	1,5690	F4/F5	True	
13	F0S1 DT	7,9177	10,8452	1,3697	F0	True	DT:
14	F0S2 DT	7,6918	10,9207	1,4198	F3	False	Front
15	F0S3 DT	7,3038	10,5891	1,4498	F3	False	surface
16	F3S1 DT	11,8859	15,3337	1,2901	F0	False	of middle
17	F3S2 DT	9,8199	12,8930	1,3129	F0	False	section
18	F3S3 DT	10,9066	15,5500	1,4257	F3	True	
19	F4S1 DT	15,7896	21,9578	1,3907	F3	False	
20	F4S2 DT	12,9032	18,9321	1,4672	F3	False	
21	F4S3 DT	9,2836	14,1572	1,5250	F4/F5	True	
22	F5S1 DT	9,7572	14,0671	1,4417	F3	False	
23	F5S2 DT	12,3533	18,9965	1,5378	F4/F5	True	
24	F5S3 DT	10,9792	17,1891	1,5656	F4/F5	True	
25	F0S1 DU	7,8667	11,6992	1,4872	F3	False	DU:
26	F0S2 DU	7,5373	11,1222	1,4756	F4	False	Front
27	F0S3 DU	7,4140	11,1486	1,5037	F4/F5	False	surface
28	F3S1 DU	13,8776	19,4875	1,4042	F3	True	of top
29	F3S2 DU	10,0506	13,5783	1,3510	F0	False	section
30	F3S3 DU	8,9027	12,2714	1,3784	F3	True	
31	F4S1 DU	12,0523	16,7271	1,3879	F3	False	
32	F4S2 DU	11,2896	15,5545	1,3778	F3	False	
33	F4S3 DU	8,8804	13,8123	1,5554	F4/F5	True	
34	F5S1 DU	8,2798	11,6025	1,4013	F3	False	
35	F5S2 DU	8,9208	12,6784	1,4212	F3	False	
36	F5S3 DU	8,4947	12,3973	1,4594	F3	False	

Table 3. The results of feature extraction and category of back surface FFB

No	Label	(I)	(σI)	(C)	Category	Results	Remarks
1	F0S1 BP	17,8477	27,6307	1,5481	F4/F5	False	BP: Rear
2	F0S2 BP	20,6675	29,5511	1,4298	F3	False	surface
3	F0S3 BP	15,7328	26,0196	1,6539	F4/F5	False	of base
4	F3S1 BP	18,8325	28,3307	1,5043	F4/F5	False	section
5	F3S2 BP	15,0107	21,5276	1,4341	F3	True	
6	F3S3 BP	15,0268	22,9386	1,5265	F4/F5	False	
7	F4S1 BP	22,7972	37,5626	1,6477	F4/F5	True	
8	F4S2 BP	22,2380	34,5376	1,5531	F4/F5	True	
9	F4S3 BP	15,0004	20,5883	1,3725	F0	False	
10	F5S1 BP	20,2776	32,8684	1,6209	F4/F5	True	
11	F5S2 BP	16,9439	26,6742	1,5743	F4/F5	True	
12	F5S3 BP	24,0895	38,0983	1,5815	F4/F5	True	

13	F0S1 BT	14,1247	24,3208	1,7219	F4/F5	False	BT: Rear surface of middle section
14	F0S2 BT	14,3170	22,0723	1,5417	F4/F5	False	
15	F0S3 BT	8,2993	11,3388	1,3662	F0	True	
16	F3S1 BT	19,2050	28,7487	1,4969	F3	True	
17	F3S2 BT	13,2111	20,1834	1,5278	F4/F5	False	
18	F3S3 BT	15,0398	20,6931	1,3759	F3	True	
19	F4S1 BT	21,7583	29,4354	1,3528	F3	False	
20	F4S2 BT	17,2836	24,7171	1,4301	F3	False	
21	F4S3 BT	12,0697	16,0695	1,3314	F0	False	
22	F5S1 BT	12,9132	19,7892	1,5325	F4/F5	True	BU: Rear surface of top section
23	F5S2 BT	13,3411	21,3701	1,6018	F4/F5	True	
24	F5S3 BT	14,4245	21,1381	1,4654	F3	False	
25	F0S1 BU	8,2519	11,5079	1,3946	F3	False	
26	F0S2 BU	7,9346	12,8586	1,6206	F4/F5	False	
27	F0S3 BU	7,6093	11,3591	1,4928	F3	False	
28	F3S1 BU	9,5628	13,6629	1,4287	F3	True	
29	F3S2 BU	11,1023	15,4696	1,3934	F3	True	
30	F3S3 BU	9,7705	13,4804	1,3797	F3	True	
31	F4S1 BU	13,1232	19,3499	1,4745	F3	False	
32	F4S2 BU	11,6777	17,0577	1,4607	F3	False	
33	F4S3 BU	10,2687	14,3380	1,3963	F3	False	
34	F5S1 BU	8,9288	12,9348	1,4487	F3	False	
35	F5S2 BU	8,6688	12,7948	1,4760	F3	False	
36	F5S3 BU	8,9799	13,6102	1,5156	F4/F5	True	

Based on the tables, the most category is found on the front surface of base section of FFB by 75%, which means that the correct image is classified is 9 of 12 images. On the middle section, the category is 42% (5 of 12) and for the top section of the category result is 25% (3 of 12). Tests performed on the back surface of base and middle section, the classification success rate is 50% (6 of 12) and at the top section of the classification results only 33.3% (4 of 12).

The highest classification results produced by the front surface of base section, because the front surface has a fruit similarity on all parts so that it has a higher scattering intensity than the other parts. This higher scattering intensity is due to a more homogeneous scattering surface than any other surface. However, the lowest classification results are produced on the front surface of top section because it has smaller fruit sizes than the rest so that small scattering areas produce low scattering intensity.

5. Conclusion

Based on the implementation of research conducted we may conclude that:

1. The highest category is obtained on the front surface of FFB at the base of 75% because the front surface has a fruit similarity on all parts so the scatter intensity is higher than the other parts.
2. The lowest category is obtained on the front surface of the ends of the FFB i.e. 25% because the size of the fruit in this section is smaller than the other parts resulting in low scattering intensity.

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